

Homework 4 Solutions

Alexander McLain

1. The Study of Assets and Health Dynamics among the Oldest Old (AHEAD) is a national longitudinal study with initial sample of 7444 respondents aged 70 years and older, and their spouses (if married). Objectives of the study include: (i) to monitor transitions in physical, functional, and cognitive health; (ii) to examine the relationship of late-life changes in physical and cognitive health to patterns of saving and income flows; (iii) to relate changes in health to economic resources and intergenerational transfers; (iv) to examine how the mix and distribution of economic, family, and program resources affect key outcomes, including institutionalization, saving, and health declines. In this problem, we look at data from the first four waves of the study, collected in 1993, 1995, 1998 and 2000. A reduced data set, `ahead.txt` or `ahead.xlsx`, is on the course website. We explore the relationship of a test of cognitive function to some physical functioning indicators. The hypothesis is that, as physical function declines, movement becomes more difficult, and hence there are fewer stimuli, leading to cognitive decline.

Here is a description of the variables contained:

```
id- subject identification number
year- study wave
sex- 1=male,2=female
age- age in years
immword- immediate word recall
delword- delayed word recall
blks- difficulty walking several blocks without help
strs- difficulty climbing flight of stairs without help
push- difficulty pulling or pushing a living room chair
bag- difficulty lifting a bag of groceries
dime- difficulty picking up a dime
iadlany- perform any instrumental activities of daily living
```

The focus of the study is to assess the association of difficulty performing instrumental activities of daily living (IADLs) to memory and cognition problems. IADLs include activities such as shopping, managing money, taking medications, using the telephone, preparing meals. They are supposed to represent the types of integrative activities that one needs to be able to perform in order to live independently in the community. The response variable here will be `iadlany`, an indicator for whether or not the subject reports difficulty with any (at least one) IADL. Before doing any analysis create the following variables:

- a sum of immediate and the delayed word recall scores as a measure of cognition (named as total word recall), and
- a time varying age variable.

Answer the following based on this data.

First, we'll read in the data. Note that the first row of data does not have the variable names. So, we'll have to use the `header = FALSE` option then fill the names in.

```
library(tidyverse)
ahead <- read.csv("ahead.csv", na.strings = "",
                 stringsAsFactors = FALSE)
str(ahead)
```

```
## 'data.frame': 26604 obs. of 12 variables:
## $ id : int 1 1 1 1 2 2 2 2 3 3 ...
## $ year : int 0 2 5 7 0 2 5 7 0 2 ...
## $ age : int 81 81 81 81 74 74 74 74 81 81 ...
## $ sex : int 2 2 2 2 2 2 2 2 2 2 ...
## $ immword: int 3 NA NA NA 4 NA NA NA 4 5 ...
## $ delword: int 1 NA NA NA 0 NA NA NA 4 4 ...
## $ blks : int 0 0 0 1 0 NA NA NA 1 1 ...
## $ strs : int 0 0 0 0 0 NA NA NA 1 0 ...
## $ push : int 1 0 0 0 0 NA NA NA 1 1 ...
## $ bag : int 0 0 0 0 0 NA NA NA 0 1 ...
## $ dime : int 0 0 0 0 0 NA NA NA 0 0 ...
## $ iadlany: int 0 1 1 1 0 NA NA NA 1 1 ...
```

```
head(ahead)
```

```
##   id year age sex immword delword blks strs push bag dime iadlany
## 1  1    0  81  2      3      1    0    0    1    0    0      0
## 2  1    2  81  2     NA     NA    0    0    0    0    0      1
## 3  1    5  81  2     NA     NA    0    0    0    0    0      1
## 4  1    7  81  2     NA     NA    1    0    0    0    0      1
## 5  2    0  74  2      4      0    0    0    0    0    0      0
## 6  2    2  74  2     NA     NA   NA   NA   NA   NA   NA     NA
```

```
ahead$totalwordrecall <- ahead$immword + ahead$delword
```

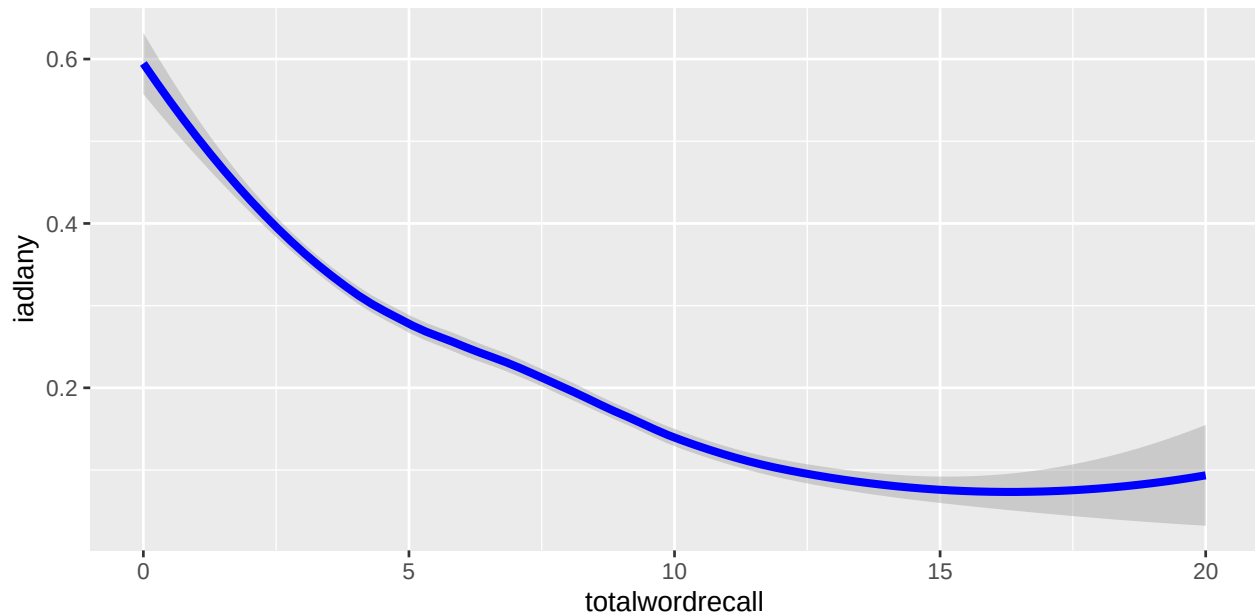
- a. Explore how difficulty with any IADL depends on age of the subject and also how it depends on total word recall. Provide tables or graphics and a brief description. (Hint: use exploratory tools, e.g. ordinary logistic regression or frequency tables.).

Here, i'm going to look at a plot of the iadlany variable by total word recall. Specifically, i'm going to plot a smoothed mean.

```
p <- ggplot(data = ahead, aes(x = totalwordrecall, y = iadlany, group = id))
p + geom_smooth(aes(group = 1), method = "loess", color = "blue", size = 1.5)
```

```
## `geom_smooth()` using formula 'y ~ x'
```

```
## Warning: Removed 7697 rows containing non-finite values (stat_smooth).
```



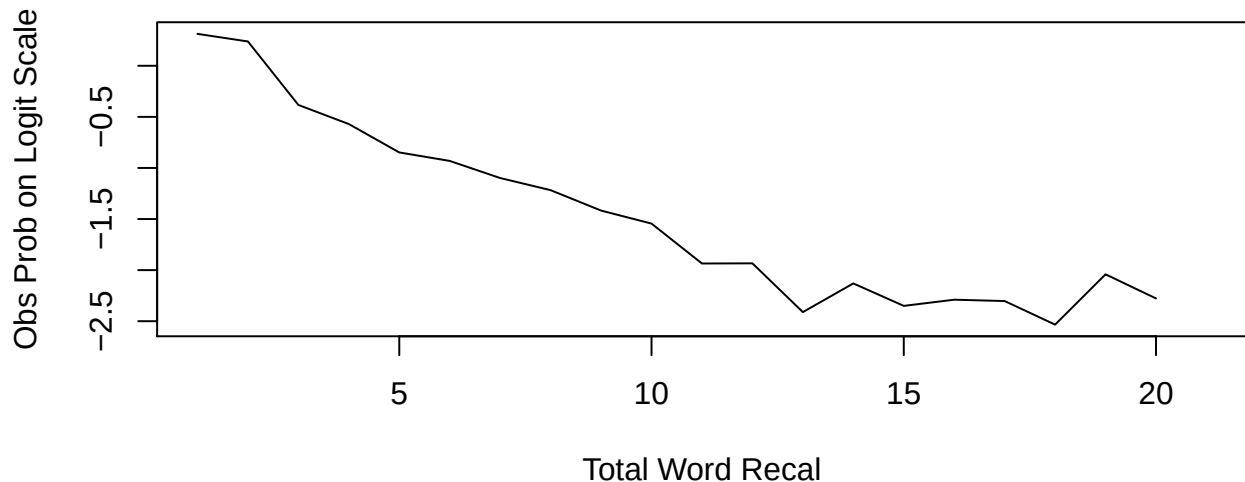
This tells us something. To look at this a little further, i'm going to:

1. get a table of the frequencies of iadlany by the word recall variable,
2. turn those into probabilities, and
3. transform the probabilities to the logit scale (since that's what we'll model things on).

```
table1 <- table( ahead$totalwordrecall, ahead$iadlany)
table2 <- data.frame( Zero_freq = as.numeric(table1[,1]),
                      One_freq = as.numeric(table1[,2]),
                      Total_freq = as.numeric(table1[,1]) +
                                as.numeric(table1[,2]))
table2$Obs_prob <- table2$One_freq/table2$Total_freq
table2$Logit_prob <- log(table2$Obs_prob/ (1 - table2$Obs_prob))
head(table2)
```

```
##   Zero_freq One_freq Total_freq  Obs_prob Logit_prob
## 1         60      82        142 0.5774648  0.3123747
## 2        160     203        363 0.5592287  0.2380322
## 3        490     334        824 0.4053398 -0.3832644
## 4        750     424       1174 0.3611584 -0.5703398
## 5        997     427       1424 0.2998596 -0.8479668
## 6       1058     417       1475 0.2827119 -0.9310494
```

```
plot( 1:21, table2$Logit_prob, type = "l", xlab = "Total Word Recal",
      ylab = "Obs Prob on Logit Scale")
```



On the logit scale, the effect appears to be mostly linear but towards the end it levels off. A quadratic effect of total word recall should work well

- b. (5 points) Now, explore the longitudinal behavior of the “iadlany” variable by relating responses at one time on a given subject to responses at other times on that same subject. Briefly describe your findings. (Hint: same as in (a)).

To do this i’m going to change the format of the data from long to wide (only focusing on the iadlany variable).

```
ahead_wide <- ahead %>% pivot_wider(id_cols = c(id, age, sex),
                                   names_from = year, values_from = c(iadlany),
                                   names_prefix = "iadlany")
head(ahead_wide)
```

```
## # A tibble: 6 x 7
##   id   age  sex iadlany0 iadlany2 iadlany5 iadlany7
##   <int> <int> <int>   <int>   <int>   <int>   <int>
## 1     1    81    2       0       1       1       1
## 2     2    74    2       0      NA      NA      NA
## 3     3    81    2       1       1       1       1
## 4     4    70    2       0       0       0       0
## 5     5    80    1       0       0       1       1
## 6     6    73    1       0       1       1      NA
```

Now let’s look at how the outcomes at one time point are related to another time point. I’m just going to use tables.

```
table(ahead_wide$iadlany0, ahead_wide$iadlany2)
```

```
##
##      0    1
## 0 3754  582
## 1  600  754
```

```
table(ahead_wide$iadlany0, ahead_wide$iadlany5)
```

```
##
##      0    1
## 0 2995  809
## 1  407  599
```

```
table(ahead_wide$iadlany0, ahead_wide$iadlany7)
```

```
##  
##           0      1  
##    0 2487  797  
##    1  318  409
```

```
table(ahead_wide$iadlany2, ahead_wide$iadlany5)
```

```
##  
##           0      1  
##    0 3125  659  
##    1  196  706
```

```
table(ahead_wide$iadlany2, ahead_wide$iadlany7)
```

```
##  
##           0      1  
##    0 2569  698  
##    1  166  460
```

```
table(ahead_wide$iadlany5, ahead_wide$iadlany7)
```

```
##  
##           0      1  
##    0 2516  456  
##    1  233  699
```

It looks like the iadlany variables at years 0 and 2, and at years 5 and 7 are the most similar to one another. As the observations get further away from each other the relationship is weaker.

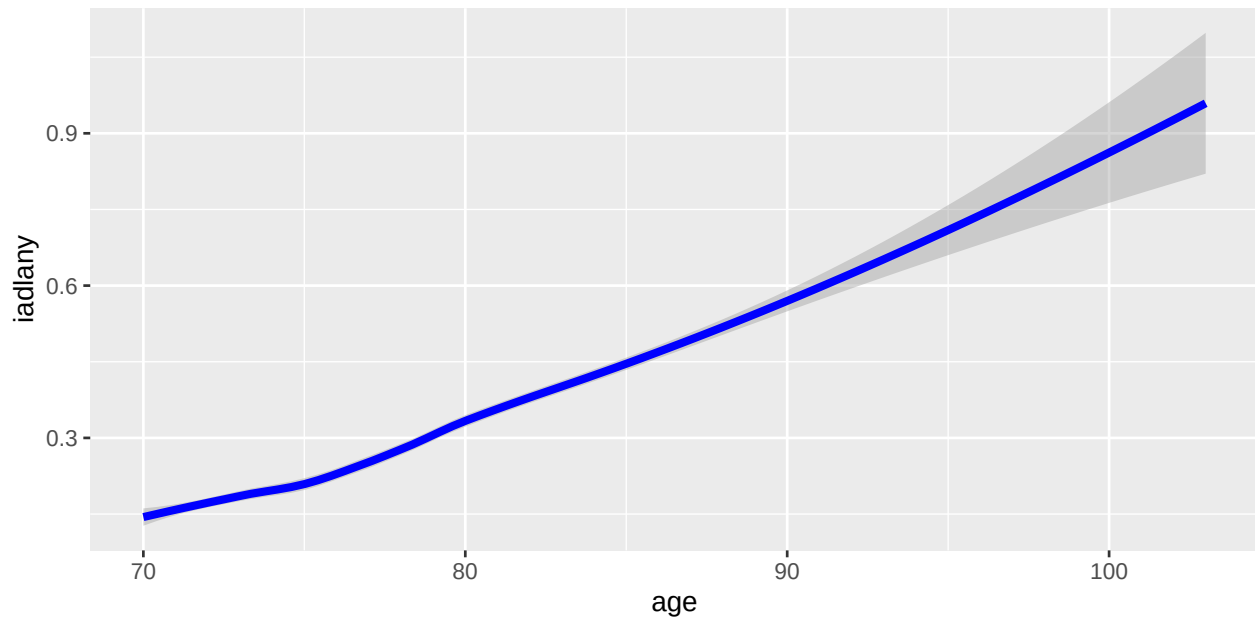
- c. (5 points) Write out a marginal or population average model for the mean response as a function of age, total word recall and sex. Specify the linear predictor and the link function.

First, I need to figure out how to include age into the model. Let's look at a plot of the outcome by age

```
p <- ggplot(data = ahead, aes(x = age, y = iadlany, group = id))  
p + geom_smooth(aes(group = 1), method = "loess", color = "blue", size = 1.5)
```

```
## `geom_smooth()` using formula 'y ~ x'
```

```
## Warning: Removed 5442 rows containing non-finite values (stat_smooth).
```



It appears that including age as linear is good. To model this we'll use the following logistic model:

$$\log \left\{ \frac{P(IADLANY_{ij} = 1)}{P(IADLANY_{ij} = 0)} \right\} = \beta_0 + \beta_1 AGE_i + \beta_2 TWR_{ij} + \beta_3 TWR_{ij}^2 + \beta_4 SEX_i$$

- d. (5 points) Building on the model in part (c), write out a "working" variance-covariance model for the vector of binary responses on a given subject. Indicate the variance function and the correlation model.

The "working" covariance model we'll use is exchangeable. Any answer is correct though. Turns out I don't really like this question.

- e. (15 points) Fit the model indicated in parts (c) and (d) using GEE with the robust estimator for the covariance of the regression coefficients. Provide an interpretation of the coefficient of total word recall and of the coefficient of sex for this model.

To fit the model with GEE we need to get rid of the missing data.

```
library(geepack)
ahead$totalwordrecall2 <- (ahead$totalwordrecall)^2
ahead <- ahead[ complete.cases(ahead), ]
form <- iadlany ~ age + totalwordrecall + totalwordrecall2 + as.factor(sex)
test <- geeglm(form, family = binomial, data = ahead, id = id,
               corstr = "exchangeable")
summary(test)
```

```
##
## Call:
## geeglm(formula = form, family = binomial, data = ahead, id = id,
##        corstr = "exchangeable")
##
## Coefficients:
##              Estimate   Std. err   Wald Pr(>|W|)
## (Intercept)   -5.791541   0.354398  267.06 < 2e-16 ***
## age            0.069060   0.004317  255.92 < 2e-16 ***
## totalwordrecall -0.186645   0.017475  114.08 < 2e-16 ***
## totalwordrecall2 0.004392   0.001070   16.87 4.01e-05 ***
```

```
## as.factor(sex)2    0.414568  0.049632  69.77  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation structure = exchangeable
## Estimated Scale Parameters:
##
##             Estimate Std.err
## (Intercept)  0.9415 0.02655
## Link = identity
##
## Estimated Correlation Parameters:
##             Estimate Std.err
## alpha      0.2896  0.0144
## Number of clusters: 6407 Maximum cluster size: 4
```

In this model we have totalwordrecall and totalwordrecall². So to give the effect of totalwordrecall we need to specify which value we're talking about. The coefficient at totalwordrecall = 0 is the main effect = -0.186645. The coefficient at totalwordrecall = 10 and 20 are

```
-0.186645 + c( 10, 20)*0.004392
```

```
## [1] -0.14272 -0.09881
```

As predicted from the plot the impact of total word recall on iadlany is decreasing as total word recall gets larger. Let's interpret the coefficient for totalwordrecall = 10:

Those with a total word recall of 10 have odds of performing instrumental activities of daily living that are 13.3% lower ($= 1 - e^{-0.14272}$) than those with a total word recall of 11, after adjusting for age and sex.

- f. (5 points) Explain in what way the estimates and inferences are valid even if the parts of the model in (d) are not correct. Obtain the estimates of the correlation model parameter(s).

The exchangeable covariance model is robust to incorrect specification in the model. Therefore even if our specification was not correct, the estimates should be valid.

- g. (5 points) Building on the model in part (e), test whether the effects of word recall varies by sex.

Let's build in some interaction terms and see if they improve things:

```
form <- iadlany ~ age + totalwordrecall*as.factor(sex) + totalwordrecall2*as.factor(sex)
test2 <- geeglm(form, family = binomial, data = ahead, id = id,
               corstr = "exchangeable")
anova(test2)
```

```
## Analysis of 'Wald statistic' Table
## Model: binomial, link: logit
## Response: iadlany
## Terms added sequentially (first to last)
##
##              Df  X2 P(>|Chi|)
## age              1 505    <2e-16 ***
## totalwordrecall  1 381    <2e-16 ***
## as.factor(sex)    1  71    <2e-16 ***
## totalwordrecall2  1  17     4e-05 ***
## totalwordrecall:as.factor(sex)  1  0      0.96
## as.factor(sex):totalwordrecall2  1  1      0.43
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

There does not appear to be any relationship with sex.

- h. **(10 points) Summarize your findings about the association of memory and cognition on difficulties with IADLs. Try to make your conclusions appropriate for a general scientific audience.**

According to our model, it appears that there is sufficient evidence to show that total word recall has a significant effect on the probability of having trouble with any IADL tasks. This relationship is diminishing as total word recall is increasing. Further, according to the model, it appears that age and gender have a significant effect on the probability of a person having trouble with any IADL tasks. However, it does not appear that sex changes the effect of total word recall on the probability that a person will have trouble with IADL skills.